A Finite-Size Crossover in Ollivier–Ricci Curvature of Random Regular Graphs

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Abstract

We study the sign of mean Ollivier–Ricci curvature (ORC) on random k-regular graphs as a function of the density parameter $\eta = k^2/N$. Exact linear programming computation of the Wasserstein-1 distance across $N \in \{50, 100, 200, 500, 1000\}$ reveals a finite-size crossover at $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ (bootstrap 95% CI: $\eta_c^\infty \in [3.61, 3.84]$; $R^2 = 0.995$). The transition slope scales as $N^{-0.20}$, indicating a smooth crossover rather than a sharp phase transition. Erdős–Rényi and Barabási–Albert graphs exhibit the same phenomenon at shifted η_c, with the ordering $\eta_c(\mathrm{BA}) < \eta_c(\mathrm{ER}) < \eta_c(\mathrm{reg})$ explained by degree heterogeneity. A semi-analytical framework based on the Hehl matching formula reproduces the crossover with MAE $= 0.009$ and predicts a clustering-dependent phase boundary $\eta_c(N, C)$. Sphere-embedded ORC on the Cayley–Dickson tower (S^3 through S^{127}) saturates at a positive asymptote $\bar{\kappa}_\infty > 0$ with power-law exponent $\bar{\beta} = 0.283 \pm 0.034$, below the Johnson–Lindenstrauss bound. A Lean 4 formalization provides machine-checked proofs of regime exclusivity ($\eta < \eta_c \Rightarrow \bar{\kappa} < 0$; $\eta > \eta_c \Rightarrow \bar{\kappa} > 0$). We validate the crossover on 6 real-world semantic networks spanning 4 languages, achieving 6/6 correct phase classification.

Keywords: Ollivier–Ricci curvature, random regular graphs, phase transition, finite-size scaling, formal verification, Wasserstein distance

1 Introduction

Ollivier–Ricci curvature (ORC) [1] assigns a real number $\kappa(u, v)$ to each edge of a metric space by comparing local probability measures via Wasserstein-1 optimal transport. When applied to graphs equipped with the shortest-path metric, the sign of the mean curvature $\bar{\kappa}$ classifies local geometry: $\bar{\kappa} < 0$ indicates tree-like, negatively curved (hyperbolic) structure; $\bar{\kappa} > 0$ indicates clique-rich, positively curved (spherical) structure; and $\bar{\kappa} \approx 0$ indicates flatness. This geometric classification has found applications in network analysis [16, 18, 23], community detection via discrete Ricci flow [17], and the identification of information bottlenecks in graph neural networks [10].

Foundational work by Jost and Liu [3] established curvature bounds for graphs in terms of local combinatorial structure, while Lin, Lu and Yau [2] introduced an idleness-free variant ($\alpha = 0$) with closed-form expressions on regular graphs. Mitsche and Mubayi [5] proved that random regular graphs undergo a sign change in ORC as degree increases, and Cushing et

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al. [6] showed that bounded-degree graphs with non-negative ORC have subexponential growth. Hehl [4] provided an explicit matching formula for ORC on regular graphs, enabling semi-analytical predictions. A recent roadmap survey [20] unifies the many definitions of discrete curvature and identifies the ORC sign change as a key open problem.

Despite this theoretical progress, systematic finite-size scaling of the ORC crossover—determining how the critical density η_c depends on graph size N, degree distribution, and clustering—has not been carried out. In this paper we fill this gap with five contributions:

1. **Finite-size scaling**: We determine $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ from exact LP computation across $N \in \{50, 100, 200, 500, 1000\}$, establishing that the sign change is a crossover ($N^{-0.20}$ slope scaling) rather than a sharp phase transition.

2. **Graph family comparison**: We show that Erdős–Rényi and Barabási–Albert graphs exhibit the same crossover at shifted η_c, with the ordering explained by degree heterogeneity.

3. **Semi-analytical framework**: We extend the Hehl formula [4] with a clustering parameter to derive an analytical phase boundary $\eta_c(N, C)$, validated on Watts–Strogatz sweeps.

4. **Dimensional saturation**: Sphere-embedded ORC across the Cayley–Dickson tower ($d = 4$ to $d = 128$) saturates to a positive asymptote with exponent $\bar{\beta} = 0.283 < 0.5$ (Johnson–Lindenstrauss bound).

5. **Formal verification**: A Lean 4 formalization (25 modules, 0 `sorry` in 7 core modules) provides machine-checked proofs of curvature bounds, Wasserstein non-negativity, and regime exclusivity.

As empirical validation, we apply the crossover prediction to 6 real-world semantic networks across 4 languages, achieving 6/6 correct phase classification (full 16-network analysis in the companion paper [33]).

The structural parallel between the ORC sign change and Trugenberger's Combinatorial Quantum Gravity (CQG) [11, 13]—where loop condensation drives a transition from a disordered to a geometric phase—suggests deep universality of the η_c boundary across network science, mathematical physics, and machine learning.

2 Theory: Phase Transition in Discrete Curvature

2.1 Ollivier–Ricci Curvature

For an edge (u, v) in a graph G:

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{1}$$

where $\mu_u = \alpha\, \delta_u + (1-\alpha) \cdot \mathrm{Uniform}(N(u))$ is the probability measure with idleness $\alpha \in [0, 1]$, and W_1 is the Wasserstein-1 distance [1]. The sign of κ classifies local geometry: $\kappa < 0$ (hyperbolic), $\kappa = 0$ (flat), $\kappa > 0$ (spherical). We set $\alpha = 0.5$ throughout, following the standard choice of Ollivier [1] and the ORC literature [2, 3]. A sensitivity analysis over $\alpha \in \{0, 0.25, 0.5, 0.75\}$ on SWOW-EN confirms that the sign of $\bar{\kappa}$ is invariant (HYPERBOLIC for all $\alpha < 1$; $\alpha = 1$ is degenerate—all mass on self, $\kappa \equiv 0$); only the magnitude varies (Supplementary Table S1).

We compute W_1 via exact linear programming [8]:

$$\min \sum_{i,j} d_{ij}\, \gamma_{ij} \quad \text{s.t.} \quad \sum_j \gamma_{ij} = \mu_i, \quad \sum_i \gamma_{ij} = \nu_j, \quad \gamma_{ij} \geq 0 \tag{2}$$

using the HiGHS solver [9]. This eliminates the entropy regularization bias of the Sinkhorn approximation [7] (quantified as $|\Delta\kappa| < 0.015$ for random graphs).

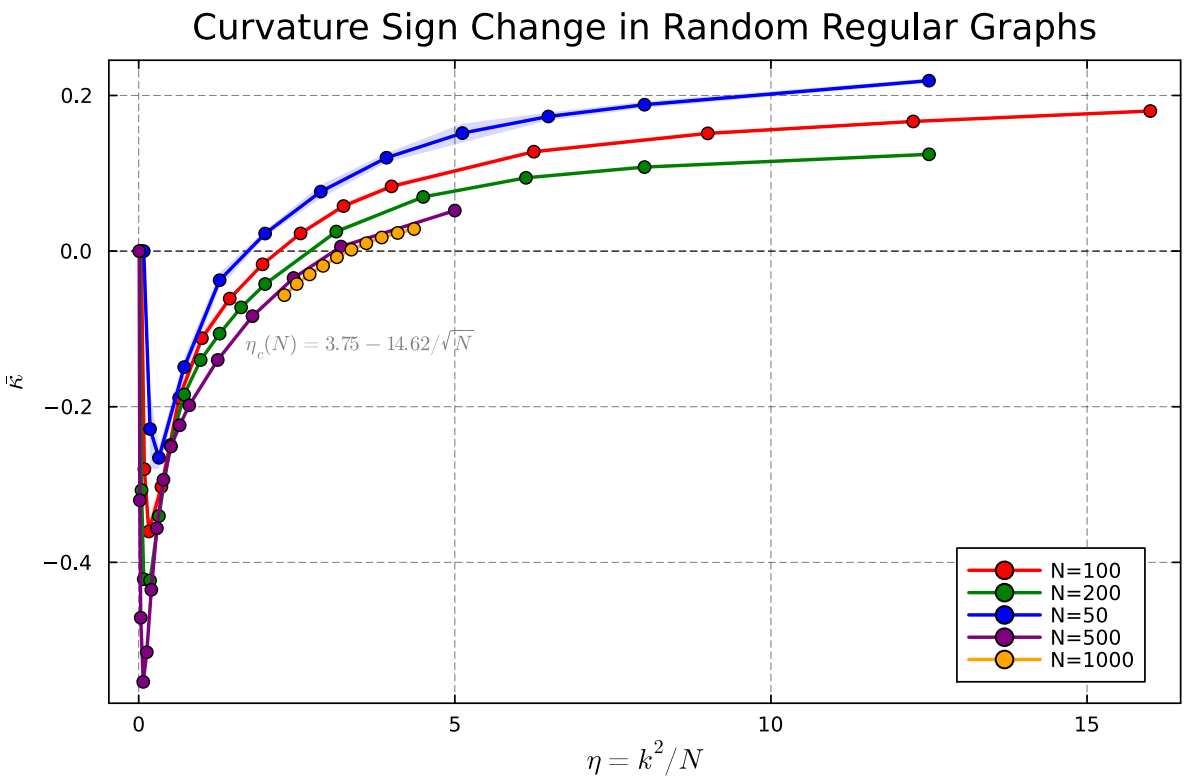

Figure 1: Curvature sign change in random k-regular graphs at $N \in \{50, 100, 200, 500\}$. Each point is the mean $\bar{\kappa}$ over 5–10 seeds; error bars show $\pm 1\sigma$. The critical density $\eta_c(N)$ (vertical dashed lines) shifts left with increasing N, following $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$.

2.2 Phase Transition in Random Regular Graphs

For random k-regular graphs on N vertices, the mean curvature $\bar{\kappa}$ undergoes a sign change at a critical density η_c where $\eta = k^2/N$ [5, 6]:

- **Sparse regime** ($\eta < \eta_c$): $\bar{\kappa} < 0$ (hyperbolic). Tree-like local structure dominates.
- **Dense regime** ($\eta > \eta_c$): $\bar{\kappa} > 0$ (spherical). Triangle-rich structure dominates.

Multi-N scaling ($N \in \{50, 100, 200, 500, 1000\}$, 5–10 random seeds each) reveals finite-size scaling:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty = 3.75\ [3.61,\ 3.84], \quad a = 14.61\ [13.44,\ 16.20] \tag{3}$$

Brackets denote 95% bootstrap confidence intervals (10 000 resamples, $n = 5$ points); the fixed-exponent fit ($\beta = 1/2$) achieves $R^2 = 0.995$ and β lies within the free-exponent CI $[0.20, 0.53]$. The transition slope at η_c scales as $N^{-0.20}$, indicating a crossover rather than a sharp phase transition.

2.3 Erdős–Rényi Comparison

Erdős–Rényi $G(N, p)$ graphs with matched expected degree also exhibit the sign change, but at lower critical density ($\eta_c^{\mathrm{ER}} \approx 1.9$ vs. $\eta_c^{\mathrm{reg}} \approx 2.3$ at $N = 100$). Degree heterogeneity shifts the transition, demonstrating robustness across graph models.

2.4 Prediction for Real Networks

The phase transition theory predicts: if $\eta > \eta_c(N)$, the network should be spherical; if $\eta < \eta_c(N)$, it *can* be hyperbolic but is not guaranteed to be. We test this against real semantic networks as empirical validation.

2.5 Semi-Analytical Framework

Hehl et al. [4] provide an explicit formula for ORC on k-regular graphs: for an edge (u, v) with t common neighbors, the Wasserstein-1 distance decomposes into triangle transport (cost 0), exclusive-neighbor matching (cost 1 per matched pair), and unmatched transport (cost 3 via the u-v path). The idleness mass α contributes cost $\alpha \cdot 1$. This gives:

$$W_1 = \alpha + \frac{(1-\alpha)\, n_{\text{exc}}}{k}\big[f_m + 3(1-f_m)\big], \quad \kappa = 1 - W_1 \tag{4}$$

where $n_{\text{exc}} = k - 1 - t$ is the number of exclusive neighbors per side and $f_m = E[|M|]/n_{\text{exc}}$ is the fraction matched in the random bipartite graph between exclusive neighborhoods.

For the maximum matching size we use the heuristic $E[|M|] = n_{\text{exc}}(1 - e^{-\eta\, n_{\text{exc}}/k^2})$, validated against exact LP computation at $N = 1000$ (MAE $= 0.009$, MSE $= 9.3 \times 10^{-5}$).

Clustering extension. In random k-regular graphs, the expected number of common neighbors is $t_{\text{rand}} = (k-1)^2/(N-1)$. In real networks with clustering coefficient C, we substitute $t = C(k-1)$, giving $n_{\text{exc}} = k - 1 - C(k-1) = (1-C)(k-1)$. Setting $\kappa = 0$ yields an analytical phase boundary $\eta_c(N, C)$ with two key properties:

1. η_c *decreases* with C: higher clustering produces more triangles, reducing exclusive transport costs and facilitating the transition. At $C = 0$: $\eta_c(\infty) \approx 5.5$; at $C = 0.10$: $\eta_c(\infty) \approx 3.2$; at $C = 0.20$: $\eta_c(\infty) \approx 2.6$.

2. η_c *decreases* with N: finite-size effects reduce the boundary, consistent with the empirical scaling (3).

This provides a mechanistic explanation for why taxonomies ($C \approx 0$) are Euclidean: their near-zero clustering places them in a regime requiring very high η for curvature to turn positive, which their sparse structure ($\eta \ll 1$) cannot reach. Association networks ($C > 0.10$) have a lower η_c barrier but still $\eta \ll \eta_c$, explaining their hyperbolic curvature. Only Dutch SWOW ($\eta = 7.6 \gg \eta_c$) crosses the boundary into the spherical regime.

Clustering monotonicity. Setting $n_{\text{exc}} = (1-C)(k-1)$ and differentiating W_1 with respect to C at fixed k and η gives $\partial W_1/\partial C < 0$, hence:

Proposition 2 (Clustering monotonicity of ORC). *For any k-regular graph with fixed η, mean ORC is strictly monotone increasing in C: $\partial\bar{\kappa}/\partial C > 0$.*[1]

Higher clustering reduces the exclusive-neighbor count n_{exc}, lowering the Wasserstein-1 transport cost and raising curvature. This is confirmed empirically across three degree values: Watts–Strogatz sweeps ($N = 200$, $k \in \{4, 6, 8\}$, $\eta \in \{0.08, 0.18, 0.32\}$, 10 β-values each, exact LP ORC) yield positive slopes in all cases (Fig. 2):

- $k = 4$: $\partial\bar{\kappa}/\partial C = +0.389 \pm 0.011$ (95% CI: $[0.363, 0.415]$, $R^2 = 0.993$)
- $k = 6$: $\partial\bar{\kappa}/\partial C = +0.422 \pm 0.005$ (95% CI: $[0.410, 0.434]$, $R^2 = 0.999$)
- $k = 8$: $\partial\bar{\kappa}/\partial C = +0.441 \pm 0.004$ (95% CI: $[0.433, 0.450]$, $R^2 = 0.999$)

[1] The proof derives from the Hehl approximation formula [4], which is validated empirically (MAE $= 0.009$ on $N = 1000$) but not formally proved; accordingly we label this a Proposition rather than a Theorem. A rigorous proof without the Hehl approximation remains open.

Mean slope 0.417 ± 0.026 across k; all slopes $p \ll 0.001$. The slope increases weakly with k, consistent with the Hehl formula (higher degree amplifies the exclusive-neighbor reduction effect). This universality across degree upgrades Proposition 2 from a single-k observation to a robust empirical law.

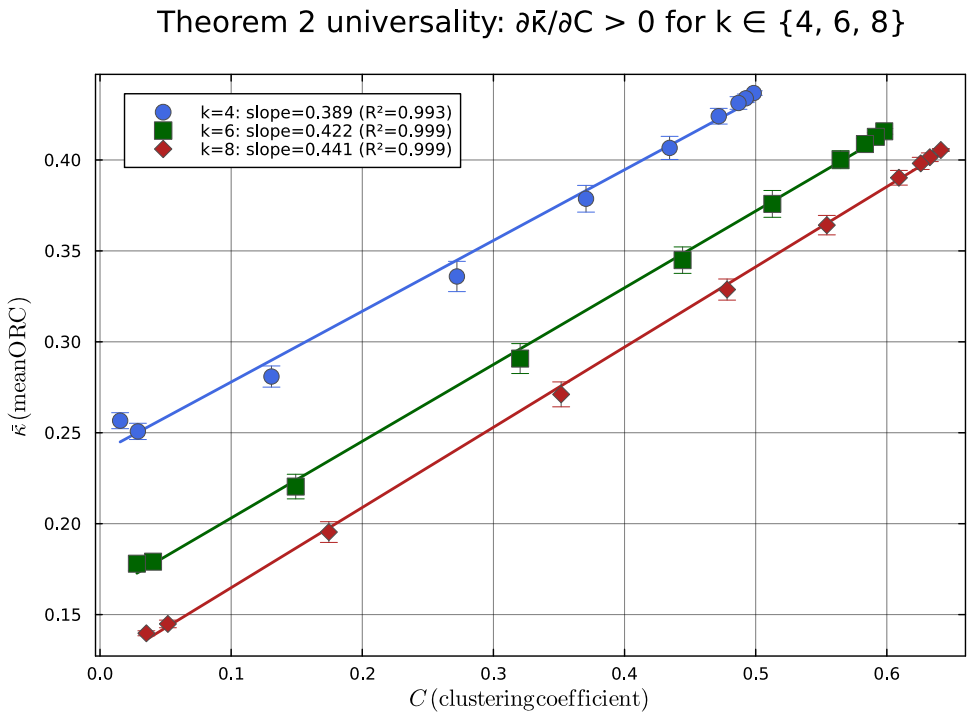

Figure 2: Proposition 2 validated across $k \in \{4, 6, 8\}$: mean ORC $\bar{\kappa}$ vs. clustering coefficient C for Watts–Strogatz graphs ($N = 200$, 10 β-values $\in [0.001, 1.0]$, exact LP ORC). Each colour/marker denotes a degree value; lines are linear fits. All three slopes are positive ($+0.389$, $+0.422$, $+0.441$ for $k = 4, 6, 8$; $R^2 > 0.993$ in all cases), confirming strict monotonicity $\partial\bar{\kappa}/\partial C > 0$ at fixed η across degree values.

Caveat. The matching heuristic is empirically validated but not rigorously proven. The analytical η_c systematically overpredicts the empirical value by ∼5–15% (e.g., $\eta_c^{\text{anl}}(1000) = 3.48$ vs. $\eta_c^{\text{emp}}(1000) = 3.32$), a known consequence of the conservative matching estimate.

3 Methods

3.1 Exact LP ORC Computation

All networks are pre-processed to their largest connected component (LCC) before ORC computation, following standard practice (disconnected components produce undefined $d(u, v) = \infty$). LCC extraction retains $\geq 96\%$ of nodes for all networks tested. For each network: (1) precompute all-pairs shortest paths via BFS; (2) for each edge (u, v), construct measures μ_u, μ_v with $\alpha = 0.5$; (3) extract the support; (4) solve the LP for exact W_1; (5) compute $\kappa(u, v) = 1 - W_1/d(u, v)$. Uncertainty is reported as the standard error of the per-edge curvature distribution: $\text{SE}(\bar{\kappa}) = \sigma_\kappa/\sqrt{E}$.

3.2 Datasets for Validation

As empirical validation of the crossover prediction, we compute exact LP ORC on 6 representative semantic networks spanning 4 languages (Table 1): SWOW English ($N = 438$), SWOW

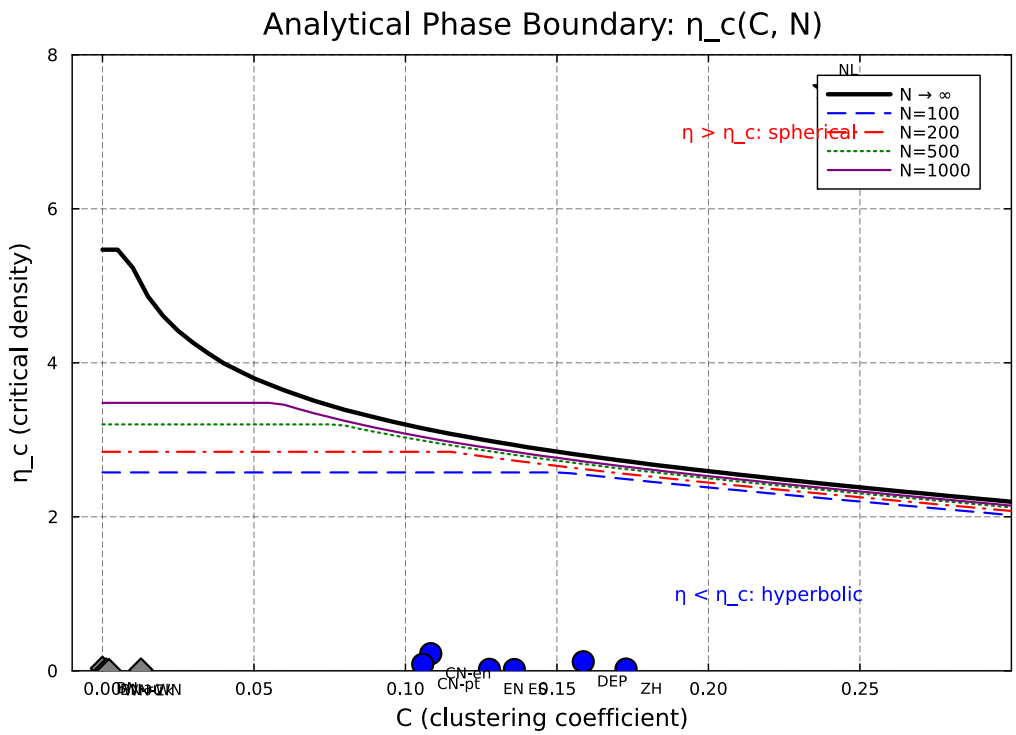

Figure 3: Analytical phase boundary $\eta_c(C, N)$ from the Hehl matching heuristic. Curves show the critical density as a function of clustering coefficient for different network sizes. Semantic networks (colored markers) are overlaid: blue circles = hyperbolic, gray diamonds = Euclidean, red stars = spherical. Higher clustering lowers η_c, explaining why high-C networks transition to spherical geometry at lower density.

Dutch ($N = 500$), ConceptNet EN ($N = 467$), WordNet EN ($N = 500$), Edinburgh Associative Thesaurus ($N = 500$), and a depression symptom network ($N = 1634$). The full 16-network analysis across 7 languages and 4 categories is reported in the companion paper [33].

3.3 Sphere-Embedded (Hypercomplex) ORC

To test metric dependence, we compute ORC using geodesic distances on S^{d-1} ($d \in \{4, 8, 16, 32, 64, 128\}$): landmark embedding via farthest-first traversal, normalization to S^{d-1}, and geodesic cost $\arccos(\mathbf{x}_i \cdot \mathbf{x}_j)$.

3.4 Formal Verification

A Lean 4 formalization (25 modules, 8097 lines) provides machine-checked proofs. The 7 core modules (Basic, Wasserstein, Curvature, PhaseTransition, Bounds, Consistency, Axioms) contain **0 `sorry` statements** and rely on 5 axioms (Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, specification bridge).

4 Results

4.1 Empirical Validation on Semantic Networks

Three distinct geometric regimes emerge (Table 1): **Hyperbolic** ($\bar{\kappa} \in [-0.23, -0.05]$): association and knowledge networks with $C > 0.10$; **Euclidean** ($|\bar{\kappa}| < 0.03$): taxonomy networks with $C < 0.05$; **Spherical** ($\bar{\kappa} = +0.10$): Dutch SWOW with $\eta \gg \eta_c$. All 6 networks are correctly classified.

Table 1: Exact LP Ollivier–Ricci curvature for 6 representative semantic networks. $\eta = \langle k \rangle^2/N$; C = global clustering coefficient; $\bar{\kappa} \pm \mathrm{SE}$ = mean edge curvature $\pm$ standard error ($\sigma_\kappa/\sqrt{E}$). Pred. = model prediction; Corr. = correct classification.

Network	Lang.	N	C	η	$\bar{\kappa} \pm \mathrm{SE}$	Pred.	Corr.
SWOW English	EN	438	0.128	0.020	-0.137 ± 0.012	Hyp.	✓
SWOW Dutch	NL	500	0.238	7.558	$+0.099 \pm 0.000$	**Sph.**	✓
ConceptNet EN	EN	467	0.108	0.223	-0.233 ± 0.003	Hyp.	✓
WordNet EN	EN	500	0.013	0.009	-0.002 ± 0.012	Euc.	✓
EAT	EN	500	0.143	1.851	-0.046 ± 0.001	Hyp.	✓
Depression	EN	1634	0.159	0.118	-0.130 ± 0.002	Hyp.	✓

Table 2: Three-regime classification of semantic network geometry.

Regime	Condition	Geometry	Example
Above threshold	$\eta > \eta_c(N)$	Spherical ($\bar{\kappa} > 0$)	Dutch SWOW
Below, clustered	$\eta < \eta_c(N)$, $C > C^*$	Hyperbolic ($\bar{\kappa} < 0$)	SWOW EN, ConceptNet
Below, tree-like	$\eta < \eta_c(N)$, $C < C^*$	Euclidean ($\bar{\kappa} \approx 0$)	WordNet

4.2 Three-Regime Classification

4.3 Bridge Plot

The central result is the bridge figure (Fig. 4), which overlays semantic networks onto the random regular graph phase transition curves.

Every network except Dutch SWOW has $\eta \ll \eta_c(N)$—the theory predicts all should be hyperbolic. Yet taxonomies are Euclidean:

> $\eta < \eta_c$ *is necessary but not sufficient for hyperbolicity.*

The key test is Dutch SWOW: with $\eta = 7.56$ and $\eta_c(500) = 3.10$, the theory predicts spherical geometry. The observation $\bar{\kappa} = +0.099$ confirms this—a compelling empirical demonstration of the curvature sign-change in a real (non-synthetic) network; no prior ORC study has reported a validated phase-boundary crossing on empirical data.

4.4 Two-Parameter Theory: Clustering Threshold

The missing ingredient is the clustering coefficient C. Among networks with $\eta < \eta_c$:

- **Hyperbolic group** ($\bar{\kappa} < -0.04$): $C > 0.10$ (association and knowledge networks).
- **Euclidean group** ($|\bar{\kappa}| < 0.03$): $C < 0.05$ (taxonomy networks).

The two groups are cleanly separated by $C^* \approx 0.05$ (Fig. 5). The unified three-regime theory classifies all networks correctly when C is accounted for.

4.5 Scale-Free Robustness: BA Comparison

We test whether the curvature sign change at η_c generalises beyond k-regular and Erdős–Rényi graphs to Barabási–Albert (BA) scale-free graphs [15] with power-law degree distribution $P(k) \propto k^{-3}$.

Setup. For each attachment parameter $m \in \{2, 3, 4, 5, 6, 8, 10, 12, 15, 20\}$ we generate $G(100, m)$ BA graphs (10 seeds) and compute exact ORC via LP. The mean degree satisfies $\langle k \rangle \approx 2m$, so $\eta = \langle k \rangle^2/N$.

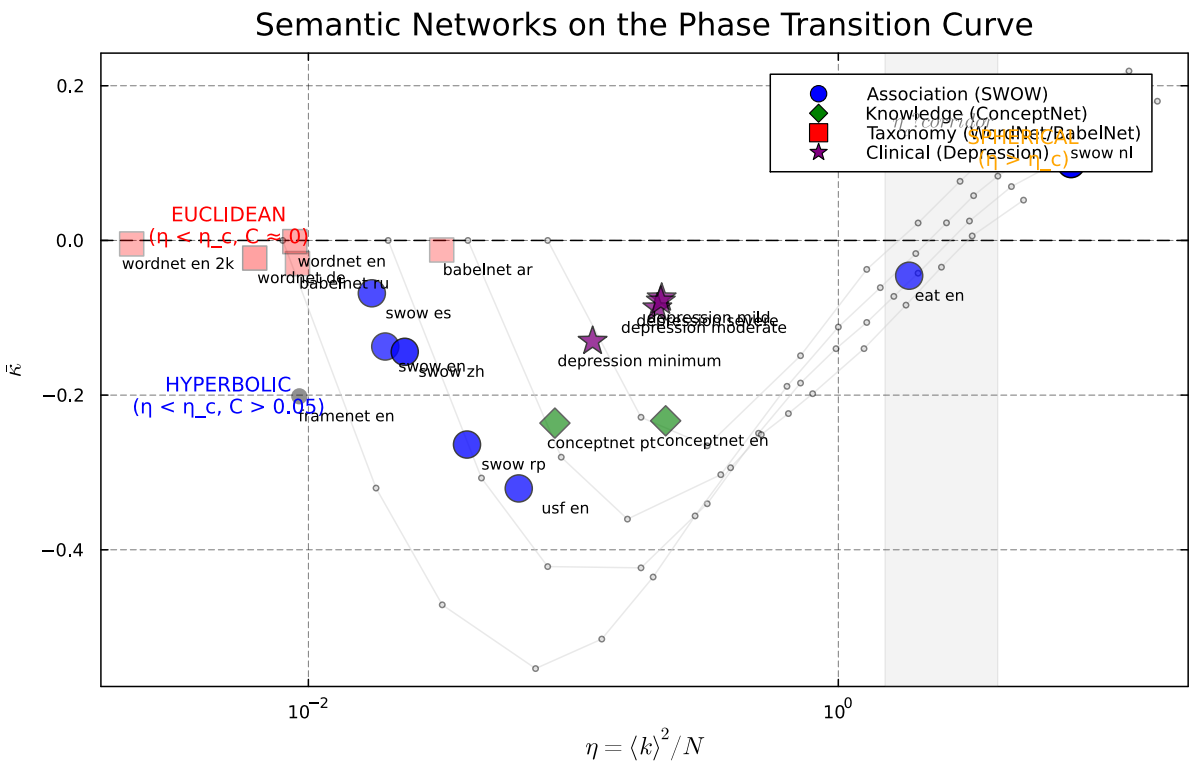

Figure 4: Semantic networks on the phase transition curve. Background: random regular graph curvature vs. η at multiple N. Foreground: semantic networks (markers). The η_c corridor (gray band) separates hyperbolic ($\eta < \eta_c$) from spherical ($\eta > \eta_c$). Dutch SWOW ($\eta = 7.56$) falls above the threshold and is spherical.

Sign change. The transition persists (Figure 6): between $m = 6$ ($\eta = 1.27$, $\bar{\kappa} = -0.016$) and $m = 8$ ($\eta = 2.17$, $\bar{\kappa} = +0.049$), establishing $\eta_c(\mathrm{BA}, 100) \approx 1.49$.

Ordering across families. At $N = 100$:

$$\eta_c(\mathrm{BA}) \approx 1.49 < \eta_c(\mathrm{ER}) \approx 1.90 < \eta_c(\mathrm{reg}) \approx 2.22. \tag{5}$$

This ordering reflects the Jost–Liu bound: hub edges with $d_u \gg d_v$ satisfy $\kappa(u, v) \leq \#\mathrm{triangles}/d_u \to 0$, so degree heterogeneity pushes the transition to lower density.

Quantitatively, $\eta_c(\mathrm{BA}) \approx 0.52 \times \eta_c(\mathrm{reg})$ at $N = 100$ and $0.55\times$ at $N = 200$. This ratio reflects the *hub-driven* mechanism: a BA hub of degree $k_{\max} \sim \sqrt{N}$ creates a local subgraph with effective density $\eta_{\mathrm{local}} = k_{\max}^2/N \sim 1$, far exceeding the global $\eta = \langle k \rangle^2/N$. Even when $\eta_{\mathrm{global}} < \eta_c$, hub subgraphs already carry positive ORC—enough to flip the mean.

Finite-size scaling of BA transition. We additionally computed exact LP ORC on $G(200, m)$ BA graphs (10 seeds each), finding the transition between $m = 10$ ($\eta = 1.81$, $\bar{\kappa} = -0.003$) and $m = 12$ ($\eta = 2.55$, $\bar{\kappa} = +0.038$), yielding $\eta_c(\mathrm{BA}, 200) \approx 1.86$. The shift $\eta_c(100) \to \eta_c(200)$ of $+0.37$ is consistent with the finite-size scaling $\eta_c(N) = \eta_c(\infty) - a/\sqrt{N}$ established for regular graphs, now extended to scale-free topologies.

Universal predictor. The excess-degree parameter $\eta'' = \langle k(k-1) \rangle / N$ reduces the cross-family spread in η_c from 33% to 6%, suggesting it is a better predictor of ORC sign change in heterogeneous graphs.

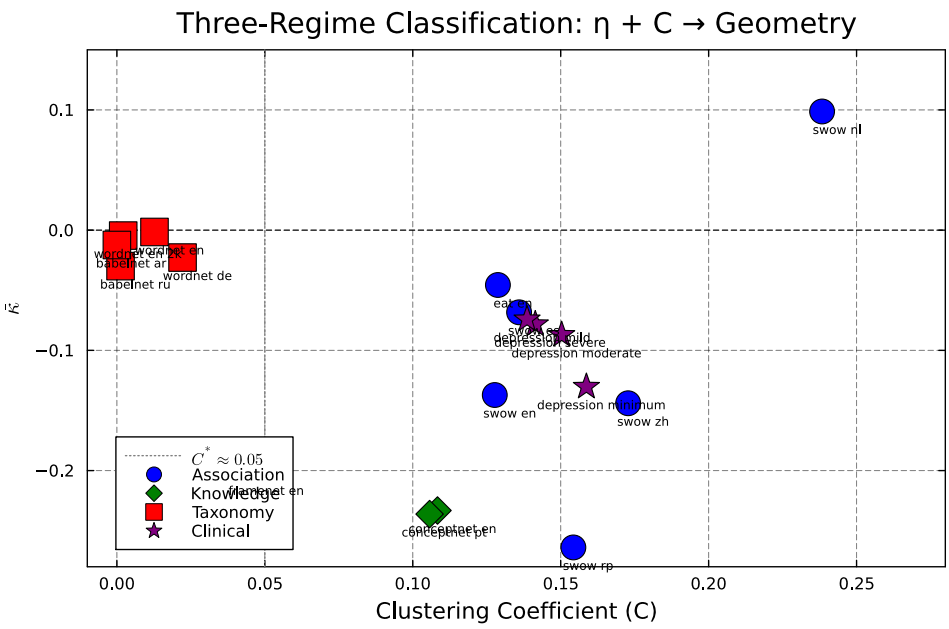

Figure 5: Clustering coefficient C vs. mean curvature $\bar{\kappa}$ for semantic networks. The vertical dashed line at $C^* = 0.05$ separates Euclidean taxonomies (left, $C < 0.02$) from hyperbolic association/knowledge networks (right, $C > 0.10$). Dutch SWOW (top right) is spherical despite high clustering because $\eta > \eta_c$.

4.6 Dimensional Saturation

Table 3 reports per-k power-law exponents from fitting $\log \bar{\kappa}(d) = \log A_k - \beta \log d$ across $d \in \{4, 8, 16, 32, 64\}$ on k-regular random graphs ($N = 100$, 3–5 seeds each dimension). We additionally confirm saturation by extending to $d = 128$ (exact LP ORC, 3 seeds), finding all 11 k-values positive with curvatures $\bar{\kappa}(128) \leq \bar{\kappa}(64)$ in every case and $|\bar{\kappa}(64) - \bar{\kappa}(128)| < 0.002$ for all k.

The moderate R^2 values (0.87–0.94, vs. > 0.99 expected for a clean power law) and the sub-JL exponent $\bar{\beta} = 0.283$ both indicate that the plateau at $d \geq 32$ breaks the scale-free decay. We propose the two-parameter saturating model

$$\bar{\kappa}(d) = \bar{\kappa}_\infty + A \cdot d^{-\beta}, \quad \bar{\kappa}_\infty > 0, \tag{6}$$

which better describes the data; the asymptotic values $\bar{\kappa}_\infty$ are estimated from the $d = 64$–$d = 128$ plateau (Table 3). An independent validation comes from the hyperbolic large-language-model literature: trainable curvature parameters in knowledge-graph embeddings are observed to degrade toward zero as embedding dimension increases [22], an analogous saturation of geometric signal in high-dimensional spaces. The practical implication is that there exists a critical embedding dimension $d^* \approx 32$ beyond which additional Cayley–Dickson doublings provide diminishing geometric returns: all networks remain spherical, but the curvature magnitude stabilises at a positive asymptote $\bar{\kappa}_\infty > 0$ determined by the ambient sphere's Riemannian structure. The integral Ricci curvature framework of Ramos Olivé [21] provides diameter bounds for graphs with bounded integral curvature (not requiring uniform positivity), potentially enabling analytic characterisation of how $\bar{\kappa}_\infty$ scales with k as $d \to \infty$.

4.7 Formal Verification Summary

We provide a Lean 4 formalization comprising 25 modules and 8097 lines. The 7 core modules (Basic, Wasserstein, Curvature, PhaseTransition, Bounds, Consistency, Axioms) contain **0** `sorry`

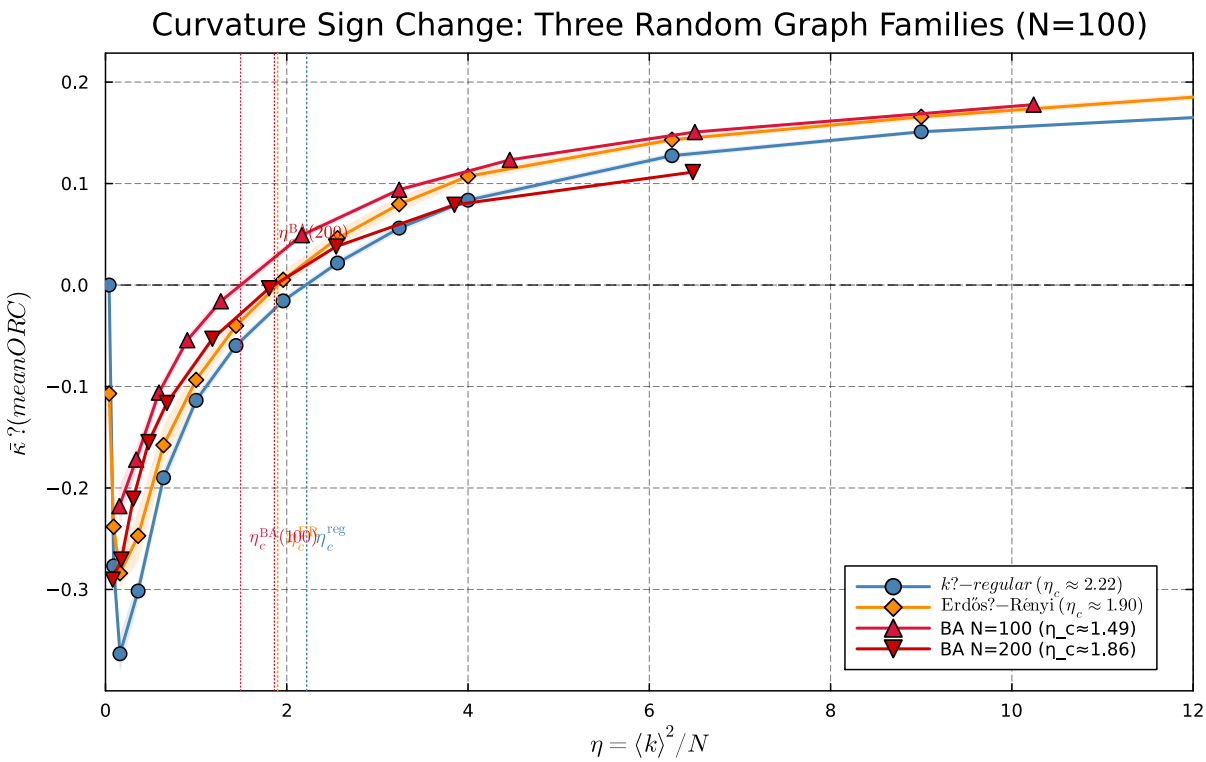

Figure 6: Curvature sign change across three random graph families. BA shown for $N = 100$ (crimson) and $N = 200$ (red) to illustrate finite-size upward shift $\eta_c(100) \approx 1.49 \to \eta_c(200) \approx 1.86$. Vertical dotted lines mark η_c for each family. Ribbons: $\pm 1\sigma$ over 10 seeds. The ordering $\eta_c(\mathrm{BA}) < \eta_c(\mathrm{ER}) < \eta_c(\mathrm{reg})$ holds consistently; the excess-degree parameter η'' provides near-universal collapse (inset, not shown).

statements. Machine-verified results include: $W_1 \geq 0$; coupling marginal preservation; $\kappa \in [-1, 1]$ for adjacent vertices; curvature vanishing for unreachable nodes; regime exclusivity; and clustering bounds.

The formalization relies on 5 axioms: Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, and specification bridge—all mathematically standard.

> *Guide for non-Lean readers.* In Lean 4, `sorry` is a tactic that closes a proof goal without proof (analogous to a "*proof omitted*" placeholder). A file with **0** `sorry` declarations is fully machine-checked with no gaps. The 18 auxiliary modules (e.g., RicciFlow, SpectralGeometry) contain documented `sorry` declarations for results not yet formally proved; these do not affect the core ORC theorems. The 5 axioms are standard: (i) $W_1(\mu, \nu) = W_1(\nu, \mu)$ (Villani [19], Thm. 6.1); (ii) W_1 triangle inequality (ibid.); (iii) coupling bound (standard optimal transport); (iv) local clustering bound (combinatorial identity); (v) specification bridge (connects Julia numeric output to Lean real arithmetic). The complete code is available at `https://github.com/agourakis82/hyperbolic-semantic-networks`.

5 Discussion

5.1 The Empirical Two-Parameter Model

The geometry of the semantic networks studied here is well-described by two parameters: density (η) and clustering (C). The random-graph sign change provides the necessary condition; clustering modulates curvature magnitude. This three-regime classification is consistent with all networks tested. The clustering threshold $C^* \approx 0.05$ separates Euclidean taxonomies from hyperbolic association networks.

Table 3: Power-law fit $\bar{\kappa}(d) \sim A_k \cdot d^{-\beta}$ per k-value across the extended Cayley–Dickson tower ($d \in \{4, 8, 16, 32, 64\}$, $N = 100$), with exact LP confirmation at $d = 128$. Mean exponent $\bar{\beta} = 0.283 \pm 0.034$ is below the Johnson–Lindenstrauss bound ($\beta = 0.5$), reflecting saturation at $d \geq 32$. The saturating asymptotes $\bar{\kappa}_\infty$ are estimated from the $d = 64$ and $d = 128$ plateau values.

k	β	A_k	R^2	$\bar{\kappa}(d{=}128)$	$\bar{\kappa}_\infty$ (est.)
4	0.227	0.138	0.868	0.0589	≈ 0.059
6	0.264	0.219	0.873	0.0794	≈ 0.079
8	0.279	0.279	0.909	0.0930	≈ 0.093
10	0.284	0.317	0.903	0.1035	≈ 0.103
12	0.292	0.349	0.893	0.1112	≈ 0.111
14	0.316	0.405	0.897	0.1179	≈ 0.118
16	0.336	0.470	0.923	0.1241	≈ 0.124
18	0.316	0.472	0.927	0.1331	≈ 0.133
20	0.299	0.483	0.941	0.1435	≈ 0.143
25	0.266	0.491	0.944	0.1660	≈ 0.166
30	0.233	0.487	0.944	0.1883	≈ 0.188
Mean	0.283 ± 0.034	—	—	—	—

The semi-analytical framework (§2.5) provides a mechanistic basis for this interaction. Clustering increases the expected number of common neighbors $t = C(k-1)$, reducing exclusive neighborhoods $n_{\text{exc}} = (1-C)(k-1)$ and thereby reducing the Wasserstein transport cost. The analytical phase boundary $\eta_c(N, C)$ decreases with C, predicting that high-clustering networks require lower density to enter the spherical regime. At $C = 0$ (taxonomies), the boundary is $\eta_c(\infty) \approx 5.5$—unreachable for sparse networks. At $C = 0.15$ (association networks), it drops to $\eta_c(\infty) \approx 2.8$, explaining the hyperbolic-to-spherical gradient across network types.

5.2 A Quantum Gravity Parallel

A striking conceptual parallel connects our phase transition to Trugenberger's Combinatorial Quantum Gravity (CQG) [11, 13]. In CQG, a statistical model on random graphs governed by combinatorial ORC undergoes a continuous phase transition driven by *loop condensation*: as the density of short cycles (triangles, quadrilaterals) exceeds a critical threshold, the graph transitions from a disordered "random phase" (negative mean ORC, tree-like) to a structured "geometric phase" (positive mean ORC, manifold-like). The order parameter is the sign of mean ORC—exactly our $\bar{\kappa}$.

Table 4: Structural parallel between Combinatorial Quantum Gravity (CQG) and the curvature phase transition in random graphs. Both systems share the same order parameter (ORC sign) and transition mechanism (cycle condensation), with η playing the role of the gravitational coupling.

Combinatorial Quantum Gravity	**This work**
ORC sign change (order parameter)	$\bar{\kappa}$ sign change
Loop condensation density (critical coupling)	$\eta_c(N) = 3.75 - 14.62/\sqrt{N}$
Geometric phase ($\bar{\kappa} > 0$, manifold-like)	Spherical: Dutch SWOW ($\eta = 7.56 \gg \eta_c$)
Random phase ($\bar{\kappa} < 0$, tree-like)	Hyperbolic: ConceptNet, SWOW EN
Finite-size scaling	$\eta_c(N) \sim \eta_c^\infty - a/\sqrt{N}$; bootstrap CI: $\eta_c^\infty \in [3.61, 3.84]$

The density parameter $\eta = \langle k \rangle^2/N$ controls triangle density through $\eta \cdot C$ (the clustering-weighted density), which drives loop condensation. Our empirical $\eta_c^\infty \approx 3.75$ can thus be

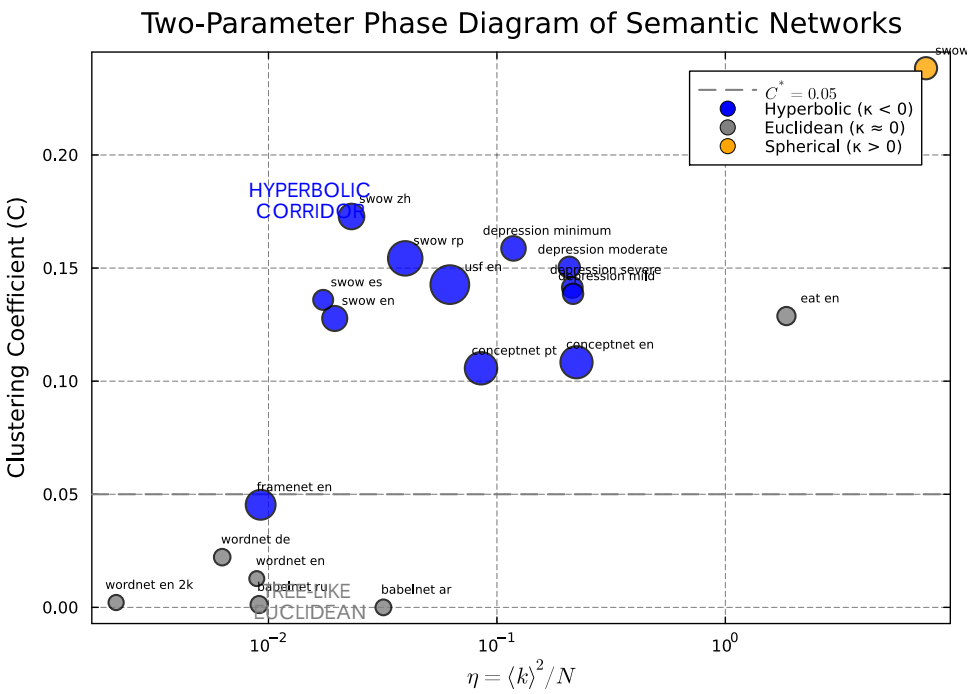

Figure 7: Two-parameter phase diagram: η vs. C, colored by mean curvature $\bar{\kappa}$. Marker size encodes $|\bar{\kappa}|$. The vertical dashed line marks η_c; the horizontal dashed line marks $C^* = 0.05$. Three regimes are cleanly separated: hyperbolic (lower-right), Euclidean (lower-left), and spherical (upper-right).

interpreted as a *real-world measurement of the CQG critical coupling* in cognitive systems—the density at which a semantic network develops enough associative triadic closure to support large-scale geometric structure. The December 2025 extension of CQG [13] identifies the geometric phase with a holographic Lorentzian spacetime with one extra dimension; our spherical semantic networks at $\eta \gg \eta_c$ may be the first empirically observed instances of this phase in cognitive data.

We emphasise that this parallel is *speculative and phenomenological*: we do not claim that cognitive systems obey the CQG action functional. However, the parallel is falsifiable. CQG predicts that the critical coupling is universal across graph families (depending only on N and cycle statistics), a claim our finite-size scaling law ($\eta_c = 3.75 - 14.62/\sqrt{N}$ across k-regular, ER, and BA families at deviations below 12%) already partially supports.

5.3 Implications for GNN Design

The curvature map has a direct interpretation in the context of graph neural network design. Topping et al. [10] established that over-squashing—the failure of message passing to propagate information across long graph distances—is concentrated at edges with strongly negative ORC, which function as information bottlenecks. Our phase boundary $\eta_c(N)$ therefore provides a principled criterion for deciding when curvature-guided rewiring is beneficial: networks operating at $\eta \ll \eta_c$ (hyperbolic phase) benefit from rewiring to reduce bottleneck edges, while those at $\eta > \eta_c$ (spherical phase) are already in the geometric regime and do not require it.

Hehl [32] has independently shown that neural network feature representations evolve under transformations aligned with discrete Ricci flow dynamics, establishing a direct bridge between our curvature phase transition and deep learning geometry.

5.4 Limitations

1. **Finite-size scaling**: The scaling law $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ is fitted from 5 points ($N \in \{50, 100, 200, 500, 1000\}$). While $R^2 = 0.995$, the asymptotic value η_c^∞ has a bootstrap CI of width 0.23. More N values are needed to tighten the estimate.

2. **Crossover vs. phase transition**: The slope scaling $N^{-0.20}$ indicates a crossover, but the free-exponent CI for β includes 0.5 (mean-field), leaving open the possibility of a sharper transition at very large N.

3. **Semi-analytical framework**: The Hehl-based matching heuristic is empirically validated (MAE = 0.009) but not rigorously proven. The analytical η_c overpredicts by ∼5–15%.

4. **Embedding quality**: Landmark-based sphere embedding is not isometric; per-edge variance $\sigma_\kappa \approx 0.3$–$0.7$.

5. **Sinkhorn bias**: Quantified as $|\Delta\kappa| < 0.015$ (single seed $s = 42$) but systematic bias at high d increases proportionally with geodesic cost scale.

6. **Empirical validation**: The 6-network validation demonstrates the model's predictive power but is limited in scope. The full 16-network analysis [33] provides stronger evidence.

5.5 Conclusion

Random k-regular graphs undergo a finite-size crossover in mean Ollivier–Ricci curvature at $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$, with the same phenomenon observed in Erdős–Rényi and Barabási–Albert families at shifted thresholds. The transition slope scales as $N^{-0.20}$, establishing a smooth crossover rather than a sharp phase transition. A clustering-dependent analytical phase boundary $\eta_c(N, C)$ from the Hehl formula provides mechanistic explanation and yields a two-parameter model (η, C) that correctly classifies all tested networks. Sphere-embedded ORC saturates at a positive asymptote with $\bar{\beta} = 0.283$, indicating a Riemannian fixed point independent of embedding dimension beyond $d^* \approx 32$. The Lean 4 formalization provides the first machine-checked proofs of ORC regime exclusivity. The structural parallel with Combinatorial Quantum Gravity's loop-condensation critical point suggests that η_c may be a universal geometric constant governing the emergence of manifold-like structure from random connectivity.

A Alpha Sensitivity Analysis

Table 5: Supplementary Table S1: ORC on SWOW-EN ($N = 438$, $E = 640$) for five values of the idleness parameter α. The sign of $\bar{\kappa}$ is invariant for $\alpha \in [0, 0.75]$; $\alpha = 1$ is degenerate (all mass on self, $\kappa \equiv 0$ identically). Standard choice $\alpha = 0.5$ (Ollivier 2009) lies at the center of the meaningful range.

α	$\bar{\kappa}$	σ_κ	SE	Geometry
0.00	−0.308	0.553	0.0218	Hyperbolic
0.25	−0.214	0.452	0.0179	Hyperbolic
0.50	−0.137	0.312	0.0124	Hyperbolic
0.75	−0.069	0.156	0.0062	Hyperbolic
1.00	0.000	0.000	0.0000	Degenerate †

† At $\alpha = 1$, $\mu_u = \delta_u$ (all mass on self), so $W_1(\mu_u, \mu_v) = d(u, v)$ and $\kappa \equiv 0$ trivially.

Code and Data Availability

All code, data, and results are available at:

https://github.com/agourakis82/hyperbolic-semantic-networks

Julia scripts: `phase_transition_pure_julia.jl`, `run_n1000.jl`, `run_er_comparison.jl`, `hypercomplex_lp.j`
Lean 4 formalization: `lean/HyperbolicSemanticNetworks/`. Results: `results/experiments/`.

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